Frohman-S-UnitCorrection: Narrow CohFT Unit 1_J Correction, Seed Family S, and Lefschetz Orthogonality Tables in FJRW Theory

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ORCID: 0009-0006-7068-3718 | GitHub: BenFrohman/Frohman-S-UnitCorrection
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Abstract

Standalone dataset and technical specification (‘docs/CORRECTION.md‘, ‘docs/BROAD_GRADING.md‘, ‘docs/UNIT_INSERTION.md‘, ‘tables/seeds.csv‘, ‘tables/not_seeds.csv‘, ‘tables/broad_grading.csv‘, ‘tables/lefschetz_orthogonality.csv‘) establishing the narrow CohFT unit 1_J selection-rule correction for the seed family S in three-block FJRW theory.

1 Scope, Scientific Honesty, and Non-Claim Boundary

Records verified selection-rule and grading corrections for the seed list S and narrow unit insertions; explicitly distinguishes closed seed tables from open Hodge/CohFT conjectures.

- **Repository:** `https://github.com/BenFrohman/Frohman-S-UnitCorrection`
- **Dedicated 1-to-1 Zenodo DOI:** `10.5281/zenodo.23652986`
- **Earliest Git Commit SHA:** `daf1c4ef200de90fa6140eb2bbe6e83081b82ecc` (2026-09-26 03:29:41 AM CDT (UTC-5))
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2 Technical Overview & Repository Specification

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Public.
Latest pass of the seed-family work. **Does not overwrite**
[BenFrohman/Frohman-MixedCorrelators-FJ](https://github.com/BenFrohman/Frohman-MixedCorrelato
FJ).
That repo stays the problem statement and the 216-cell ledger.
The CohFT unit is the generator of the **narrow** \{}(J\{})-sector:
$$
\{}mathbf{1}_J\{}in H_J,\{}qquad \{}mathrm{Fix}(J)=\{}{0\{}},\{}qquad \{}deg(\{}mathbf{1}_J)
$$

It is **not** the Jacobian degree-0 line \{}(R_0\{}simeq H^{4,0}\{}) in the identity sector.
The broad space is exactly
$$
\{}mathrm{Jac}(F)^{\{}langle J\{}rangle},\{}qquad \{}dim=2605,
$$
Hilbert \{}((1,426,1751,426,1)\{}). There is no \{}(2604\{}) complement.
Seed family, still OPEN:

3 Cryptographic Provenance & File Inventory

Every source artifact in `BenFrohman/Frohman-S-UnitCorrection` is audited via `titan-audit` (`audit/TITAN_AUDIT_CERTIFICATE.json`) and sealed under a deterministic SHA-256 Merkle tree via `sealguard` (`PROVENANCE_MERKLE_ROOT.json`) signed with Ed25519 key fingerprint `b428fb72676f1c52`.

Repository Path	Size	SHA-256 Prefix
`.github/workflows/docs.yml`	1109 B	`c3cf79ee1713c18c...`
`.github/workflows/release.yml`	571 B	`d28204063d8f1334...`
`LICENSE`	11355 B	`9da72c2bee4f20ba...`
`NOTICE`	336 B	`f7a8c9f6feae3436...`
`PARENT.md`	470 B	`88f2c459d19f66b7...`
`README.md`	1662 B	`25544aec329a5bff...`
`docs/UNIT_CORRECTION.md`	1278 B	`732b7c7c59a1ff09...`
`docs/WORKFLOWS.md`	697 B	`0d04b463cea1d015...`
`tables/broad_grading.csv`	297 B	`6cc9470e2c190ad4...`
`tables/lefschetz_orthogonality.csv`	254 B	`756b17e210b19b92...`
`tables/not_seeds.csv`	868 B	`58962daeefcd530e...`
`tables/seeds.csv`	915 B	`99d5ffd0188453f8...`

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